

Influence of Background Noise on the Temporal Perception of Vibrotactile Stimuli

Takumi Kuhara¹[0009–0002–3966–2346] and Yoshihiro Tanaka^{1,2}[0000–0001–7917–1379]

¹ Nagoya Institute of Technology, Nagoya, Japan
t.kuhara.538@nitech.jp

² Inamori Research Institute for Science, Kyoto, Japan

Abstract. Changes in perception induced by external factors provide insights into underlying perceptual mechanisms and potential applications. This study investigates the mechanisms of temporal perception for vibrotactile stimuli, specifically focusing on the minimum inter-stimulus interval (ISI) required to perceive two square wave pulses as discrete events. Psychophysical experiments were conducted under four background noise levels and three pulse durations that evoke different intensity perception. Subsequently, skin-propagating vibrations were measured using an accelerometer to dissociate physical mechanical interference from higher-level neural processing. The results revealed that the perceptual threshold for ISI significantly increases as the background noise intensity rises, and statistical analysis showed a significant main effect of noise level, but not of pulse duration. These findings suggest that vibrotactile events are perceived as temporal “chunks” and we hypothesized that we use information governed by an energy integration process rather than instantaneous events for vibrotactile perception.

Keywords: Vibrotactile, Temporal Structure, Temporal masking

1 Introduction

Changes in perception induced by external factors provide insights into underlying perceptual mechanisms and potential applications. Masking effects and adaptation in physical sensations can be leveraged for data reduction and perceptual modulation. Prior studies have shown that these phenomena can reduce the detection sensitivity and intensity perceived in vibrotactile stimulation [1–3]. Whereas these prior studies focused on the amplitude, the temporal change can also be critical for vibrotactile perception since people are capable of perceiving the amplitude transitions over time.

Thus, for investigating the temporal perception for vibrotactile stimuli, we measured the minimum Inter-stimulus interval (ISI) for successive pulses to be perceived as individual events under multiple noise conditions, which were presented in the background of the stimuli. Additionally the pulses were set to an amplitude that was ensured to be perceivable, and also conducted the experiment with conditions that lengthened the duration of the pulses for stronger perceived

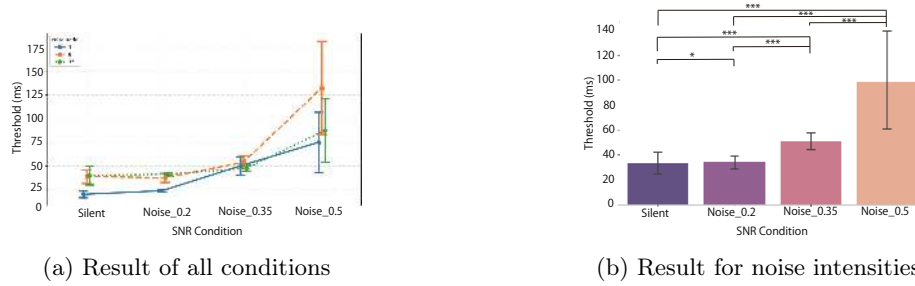


Fig. 1: The temporal threshold of the square waves; the left and right panels show the results of all conditions and those in noise intensities, respectively.

intensity [4, 5]. We analyze skin-propagating vibrations alongside psychophysical data to dissociate physical mechanics from higher-level neural processing in vibrotactile perception, and discussed the mechanism of temporal perception on the basis of the results.

2 Methods

Fourteen male participants (aged 21–24) performed a psychophysical task with written informed consent, following approval from the institutional ethics committee (2021-8). Stimuli were delivered to the index finger, and skin vibrations were captured via a ring-mounted accelerometer (Model 2302B, Showa Sokki) placed between the first and second joints to compare ISI thresholds with physical skin mechanics. We employed a 4 (noise levels: 0, 0.20, 0.35, 0.50) $\times$ 3 (pulse durations: 1, 8, 15 cycles of 2 ms square waves) within-subjects design. Pink noise was adopted for background noises, as pink noise had the most similar spectral distribution to skin propagating vibration based on the skin mechanical properties [6]. The pulse amplitude was fixed at 0.8 (273 m/s^2), which participants could detect in preliminary experiments. Variable pulse durations were used to investigate the effect of stronger perceived intensity for pulse detection within the $<30 \text{ ms}$ "pulse event" range [5]. The ISI threshold for perceiving two pulses as distinct events was determined using the method of limits. The ascending and descending conditions (5 ms steps from 5 to 200 ms) were each conducted twice, in a random order, and the average time interval threshold for the participants was calculated. For statistical analysis, an Aligned Rank Transform (ART) two-way repeated measures ANOVA [7] was applied due to the non-normal distribution of the data, followed by Wilcoxon signed-rank tests with Bonferroni correction for post-hoc comparisons.

3 Results

The ART two-way repeated measures ANOVA revealed a significant main effect of noise level ($F(3, 39) = 83.5, p < 0.001$) but neither for pulse duration ($p = 0.064$) nor the interaction ($p = 0.107$). Fig. 1a shows that the participants' responses to the time interval between the two pulses varied depending

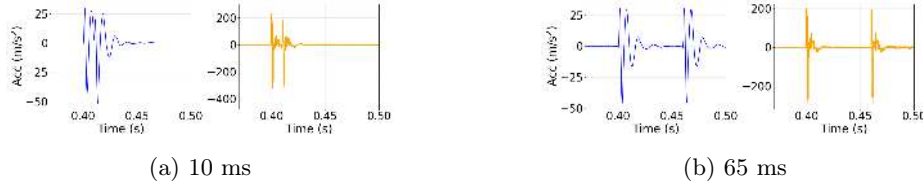


Fig. 2: The acceleration measured for the silent condition. The blue and orange plots show the acceleration of skin propagating vibrations and the vibrotactile actuator, respectively.

on the noise level, but not as clearly with the pulse duration. Fig. 1b shows that when there was background noise, the mean ISI threshold increased significantly along with the noise intensity. Fig. 2 shows the measured skin vibrations of one participant for the silent condition with the intervals of 10 and 60 ms as a representative example. A critical finding was the contrast between physical vibration and perception. Even when the acceleration of skin vibrations from the first pulse had decayed before the onset of the second pulse, participants perceived it as a unified event.

4 Discussion

The results demonstrated that the noise presented before the second pulse stimulus critically affects the detection of that pulse, suggesting that the human tactile system perceives vibrotactile inputs as temporal “chunks”, and we cumulate multiple chunks for tactile perception. Thus, this observation implies that vibrotactile perception is driven by integrated instantaneous energy that is continuously updated by incoming stimuli and decays over time. In noisy conditions, the baseline internal energy remains elevated. Consequently, even if a pulse’s instantaneous SNR exceeds the detection threshold, its contrast relative to the elevated baseline may be insufficient for the brain to register a second distinct event, requiring a longer ISI for detection of the second pulse.

Future work will focus on developing a quantitative computational model to validate these dynamics across diverse transition patterns and individual sensitivity profiles.

Open Research Questions and Interdisciplinary Exchange

Our findings suggest that vibrotactile stimuli are perceived as temporal chunks integrated into an internal energy state. However, the specific neural substrates and firing patterns underpinning this integration remain open questions. We invite **neuroscience and cognitive psychology** researchers to discuss how these perceptual windows correlate with cortical neural integration times. [If collaboration is sought: Furthermore, we are seeking collaborators with expertise in **EEG/neural modeling** to investigate hypothesized internal energy state.]

Acknowledgment

This work was supported by JST SPRING Grant Number JPMJSP2112, and Inamori Research Institute for Science.

References

1. Verrillo, Ronald & Gescheider, George & Calman, Bruce & Doren, Clayton. (1983). Vibrotactile masking: Effects of oneand two-site stimulation. *Perception & Psychophysics*. 33. 379-387. 10.3758/BF03205886.
2. Gescheider GA, O'Malley MJ, Verrillo RT. Vibrotactile forward masking: evidence for channel independence. *J Acoust Soc Am*. 1983 Aug;74(2):474-85. doi: 10.1121/1.389813.
3. Goble AK, Hollins M. Vibrotactile adaptation enhances amplitude discrimination. *J Acoust Soc Am*. 1993 Jan;93(1):418-24. doi: 10.1121/1.405621. PMID: 8423258.
4. Kuhara, Takumi & Yukawa, Hikari & Tanaka, Yoshihiro. (2025). Influence of Longterm Duration and Damping Shapes to Perceived Intensity for Vibrotactile Stimulation., *Proceedings in IEEE/SICE International Symposium on System Integration*, 930-935. 10.1109/SII59315.2025.10870919.
5. Remache-Vinueza, B., Trujillo-León, A. & Vidal-Verdú, F. Vibrotactile stimulus duration threshold for perception of pulse to vibration transition. *Sci Rep* 15, 5057 (2025). <https://doi.org/10.1038/s41598-025-85778-6>
6. Kuhara, Takumi & Kawai, Akifumi & Inoue, Yuto & Yukawa, Hikari & Tanaka, Yoshihiro. (2026). Exploring Perceptual Effects of Phase Spectra in Vibrotactile Rendering., *Proceedings in IEEE/SICE International Symposium on System Integration*, 116-121. 10.1109/SII64115.2026.11404399.
7. Jacob O. Wobbrock, Leah Findlater, Darren Gergle, and James J. Higgins. 2011. The aligned rank transform for nonparametric factorial analyses using only anova procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. Association for Computing Machinery, New York, NY, USA, 143–146. <https://doi.org/10.1145/1978942.1978963>